THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · NCERT-TYPE MCQ TEST SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Muscles in the walls of the intestine are: 1

(a) striated muscles **(b)** cardiac muscles
(c) smooth (involuntary) muscles **(d)** voluntary muscles

2. Frogs and toads belong to: 1

(a) Reptilia **(b)** Amphibia **(c)** Aves **(d)** Pisces

3. The ova are produced in the: 1

(a) uterus **(b)** ovaries **(c)** testes **(d)** vagina

4. A biogeochemical cycle is the: 1

(a) cycle of day and night
(b) cycle of the Moon
(c) cycling of nutrients through living and non-living parts of the Earth
(d) cycle of seasons

5. An example of a chemical change is: 1

(a) burning of magnesium **(b)** tearing paper
(c) dissolving sugar **(d)** melting of ice

6. Alloys are: 1

(a) heterogeneous **(b)** compounds
(c) homogeneous mixtures of metals **(d)** pure metals

7. The formula of calcium chloride is: 1

(a) Ca_2Cl_2 **(b)** Ca_2Cl **(c)** $CaCl$ **(d)** $CaCl_2$

8. Isotopes are atoms of the same element with the same: 1

(a) mass number but different atomic numbers **(b)** atomic number but different mass numbers
(c) number of nucleons **(d)** number of neutrons

9. A ball falls from rest for 2 s. Its velocity (g = 10 m/s squared) is: 1

(a) 5 m/s **(b)** 20 m/s **(c)** 10 m/s **(d)** 40 m/s

MASTERMIND

10. A fielder draws his hands back while catching a fast ball to: 1

(a) increase the force
(b) reduce the mass
(c) increase the time of stopping and reduce the force
(d) stop the ball faster

11. A lever is a rigid bar that rotates about a fixed point called the: 1

(a) effort **(b)** load **(c)** fulcrum **(d)** axle

12. SONAR stands for: 1

(a) sound navigation and ranging **(b)** sound night and ranging
(c) ship navigation and radar **(d)** sound network and range

13. Chromosomes are made of: 1

(a) DNA and proteins **(b)** only fat **(c)** only starch **(d)** only water

14. Grass regrows after being cut because of the: 1

(a) lateral meristem **(b)** apical meristem
(c) intercalary meristem **(d)** cork cambium

15. The binomial system of nomenclature was introduced by: 1

(a) Aristotle **(b)** Darwin **(c)** Carolus Linnaeus **(d)** Mendel

16. Yeast reproduces mainly by: 1

(a) binary fission **(b)** budding **(c)** fragmentation **(d)** spores only

17. All water on the Earth, in oceans, rivers, ice and groundwater, forms the: 1

(a) hydrosphere **(b)** biosphere **(c)** atmosphere **(d)** geosphere

18. A well-tested scientific theory: 1

(a) can never be changed
(b) explains many observations and makes testable predictions
(c) is just a guess
(d) is a measurement

19. A mixture of sand and iron filings is: 1

(a) homogeneous **(b)** a pure substance **(c)** a compound **(d)** heterogeneous

20. 12 g of carbon burns completely in 32 g of oxygen. The mass of carbon dioxide formed is: 1

(a) 20 g **(b)** 44 g **(c)** 12 g **(d)** 28 g

21. The neutron was discovered by: 1

(a) Bohr **(b)** James Chadwick **(c)** Rutherford **(d)** Thomson

22. Speed is a: 1

(a) force **(b)** scalar quantity **(c)** unit of length **(d)** vector quantity

23. 1 newton is the force that gives an acceleration of 1 m/s squared to a mass of: 1

(a) 10 kg **(b)** 1 kg **(c)** 1 g **(d)** 100 g

24. A force of 10 N moves a body through 5 m in the direction of the force. The work done is: 1

(a) 50 J **(b)** 2 J **(c)** 5 J **(d)** 15 J

25. A region in a sound wave where the particles are crowded together is a: 1

(a) compression **(b)** rarefaction **(c)** trough **(d)** crest

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26. Organisms made of a single cell are called: 1

(a) unicellular **(b)** non-living **(c)** multicellular **(d)** acellular

27. Food is transported to different parts of the plant by the: 1

(a) xylem **(b)** epidermis **(c)** cork **(d)** phloem

28. Plants with vascular tissues but no seeds are: 1

(a) angiosperms **(b)** gymnosperms **(c)** bryophytes **(d)** pteridophytes

29. The anther produces: 1

(a) pollen grains **(b)** nectar **(c)** ovules **(d)** seeds

30. The polar regions are colder than the equator mainly because: 1

(a) the Sun does not shine
(b) they have no atmosphere
(c) sunlight falls obliquely and is spread over a larger area
(d) they are nearer to the Sun

31. An aircraft ran out of fuel because of a mix-up of units, showing that: 1

(a) units do not matter
(b) fuel is not measurable
(c) planes need no fuel
(d) units must be used carefully and consistently

32. The separation of cream from milk is done by: 1

(a) sublimation **(b)** filtration **(c)** centrifugation **(d)** distillation

33. The molecular mass of CO_2 is (C = 12, O = 16): 1

(a) 56 u **(b)** 44 u **(c)** 28 u **(d)** 32 u

34. The number of neutrons in $^{35}_{17}\mathrm{Cl}$ is: 1

(a) 52 **(b)** 35 **(c)** 18 **(d)** 17

35. A straight line parallel to the time axis on a velocity-time graph indicates: 1

(a) uniform velocity **(b)** increasing acceleration
(c) retardation **(d)** rest only

36. Friction always acts: 1

(a) at right angles to motion
(b) downward
(c) opposite to the direction of motion or tendency of motion
(d) along the direction of motion

37. The SI unit of power is the: 1

(a) joule **(b)** kg **(c)** watt **(d)** newton

38. The audible range of frequencies for humans is about: 1

(a) 200 Hz to 2000 Hz **(b)** 20 kHz to 200 kHz
(c) 2 Hz to 200 Hz **(d)** 20 Hz to 20000 Hz

39. The organelle that packages and dispatches materials in vesicles is the: 1

(a) ribosome **(b)** nucleus **(c)** mitochondrion **(d)** Golgi apparatus

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40. The epithelium of the intestine is: 1

(a) columnar **(b)** squamous
(c) stratified squamous **(d)** cuboidal

41. Flatworms such as the tapeworm belong to: 1

(a) Nematoda **(b)** Mollusca **(c)** Platyhelminthes **(d)** Annelida

42. The cell formed by fusion of gametes is the: 1

(a) gamete **(b)** zygote **(c)** seed **(d)** spore

43. The chemicals that caused the ozone hole are: 1

(a) methane **(b)** chlorofluorocarbons (CFCs)
(c) nitrogen **(d)** carbon dioxide

44. Models of the atom have changed over time because: 1

(a) the atom changed **(b)** old models were hidden
(c) scientists were bored **(d)** new evidence required changes

45. A mixture of chalk powder in water can be separated by: 1

(a) filtration **(b)** sublimation **(c)** chromatography **(d)** evaporation

46. The formula of aluminium oxide is: 1

(a) AlO **(b)** Al_2O_3 **(c)** AlO_3 **(d)** Al_3O_2

47. Isobars are atoms with the same: 1

(a) atomic number **(b)** chemical properties
(c) mass number but different atomic numbers **(d)** number of neutrons

48. A ball is thrown vertically upward. At the highest point its velocity is: 1

(a) zero **(b)** 10 m/s **(c)** maximum **(d)** equal to initial

49. Seat belts are used to: 1

(a) stop passengers from moving forward due to inertia
(b) reduce friction
(c) increase speed
(d) make the car heavier

50. In a lever of the first class, the fulcrum is: 1

(a) absent **(b)** at the end next to effort
(c) between the load and the effort **(d)** at the end next to load

END OF QUESTION PAPER **Total: 50 Marks**

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